Strength-two trades and transversal cubic gates: the Clebsch cubic-phase codes and their magic

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Abstract

Can a symmetric point configuration define a multiqudit phase resource whose structure and preparation cost can both be certified? We study this question through strength-two signed trades: cancellation of their quadratic moments makes the third moment a transversal logical phase. Conic matchings give error-detecting codes with binary-invariant logical cubics, while an elementary translation trade supplies the same $[[2p, p-1, 2]]_p$ parameters at every prime $p \geq 5$. Exact Hessian-rank censuses identify the conic resources through their Pauli spectra. The ten-qudit example over $\mathbb{F}_{11}$ is not Clifford-equivalent to a product across any bipartition. For the six-qudit example over $\mathbb{F}_7$, joint preparation beats a specified independent primitive-distillation menu at the same target block infidelity: fewer than 16.116 raw inputs versus a lower bound of 54, or 36 when all weighted cubic types have unit cost. This comparison assumes independent uniform Z noise, ideal Clifford operations, and input error at most 1%. A chordal restriction of the logical phase explains how geometric sign choices act as gate inversion and changes of logical frame.

1 Signed trades as quantum codes

A transversal gate acts separately at each physical site, yet it can induce a phase coupling many logical qudits. Such a gate can prepare a whole multiqudit resource in one error-detecting block. This raises two questions: how can the resulting resource be distinguished from independently prepared cubic states, and when does joint preparation save raw resources at a common output error? We use finite geometry to choose the logical phase and exact Pauli data to answer these questions for two small codes.

The examples give different answers. The $[[22, 10, 2]]_{11}$ code prepares a state that cannot become a product across any bipartition, even after an entangling Clifford change of coordinates (Corollary 2.4). For the $[[14, 6, 2]]_7$ code, Theorem 3.2 proves a preparation advantage over a specified separate-distillation menu. The same product-exclusion test does not settle that six-qudit state. These are high-rate error-detecting resource factories; distance two is used to reject errors, not to correct an arbitrary single-site fault.

The common mechanism is simple. Two signed point sets agree on all quadratics, so a cubic phase cannot see the stabilizer coordinate. Its surviving third moment becomes the logical phase. This is the signed-orthogonality mechanism of qudit transversal-gate constructions [1, 3, 4], expressed below for an arbitrary coupled cubic in Theorem 1.1. Our contributions are the resource interpretation of the conic phases, their exact spectra and product consequences, and the bounded preparation comparison. The translation example shows why the code parameters alone do not distinguish the geometry.

Fix a prime $p \geq 5$ and write $\omega = \exp(2\pi i/p)$. A qudit has basis $\{|x\rangle : x \in \mathbb{F}_p\}$, with $X(a)|x\rangle = |x+a\rangle$ and $Z(b)|x\rangle = \omega^{bx}|x\rangle$. A *strength-two signed trade* is a list of distinct points $x_i \in \mathbb{F}_p^k$, with signs $\epsilon_i \in \{1, -1\}$, such that

$$\sum_i \epsilon_i = 0, \qquad \sum_i \epsilon_i x_i = 0, \qquad \sum_i \epsilon_i x_i x_i^T = 0. \tag{1}$$

These equalities are in $\mathbb{F}_p$. Let E have rows x_i^T and let $L = \{a\mathbf{1} + Eu : a \in \mathbb{F}_p,\ u \in \mathbb{F}_p^k\} \subseteq \mathbb{F}_p^n$. The coordinatewise product span is denoted by $L^{\circ 2}$. A Calderbank–Shor–Steane (CSS) code is specified here by commuting X- and Z-stabilizer spaces: vectors in those spaces prescribe products of the corresponding Pauli operators whose common $+1$ eigenspace is the code. The notation $[[n, k, d]]_p$ means n physical qudits, k logical qudits and distance d.

Theorem 1.1 (Trade-to-code dictionary)**.** *Suppose $n = 2p$, $k = p - 1$, and $\dim L = p$. Then the CSS code with X-stabilizer space $S = \langle \mathbf{1} \rangle$ and Z-stabilizer space $L^\perp$ has parameters $[[2p, p-1, 2]]_p$. The signed transversal operation $\bigotimes_i M_{\epsilon_i}$, where $M_c|z\rangle = \omega^{cz^3}|z\rangle$, induces the logical diagonal gate*

$$U_F|u\rangle = \omega^{F(u)}|u\rangle, \qquad F(u) = \sum_i \epsilon_i (x_i^T u)^3. \tag{2}$$

More generally, for $S \subseteq L$, a physical cubic with weights w_i is constant on every coset of S in L if and only if $\sum_i w_i s_i b_i c_i = 0$ for all $s \in S$ and $b, c \in L$. For $S = \langle \mathbf{1} \rangle$ this condition is $w \perp L^{\circ 2}$.

Proof. Put $D = \operatorname{diag}(\epsilon_i)$. Equation (1) says that L is isotropic for the nondegenerate pairing $v^T D w$. Its dimension is half the length, so $L^\perp = DL$. In particular $S \subseteq L$, and the CSS dimension is $\dim L - \dim S = p - 1$. An encoded basis is

$$|u_L\rangle = p^{-1/2} \sum_{a \in \mathbb{F}_p} |a\mathbf{1} + Eu\rangle.$$

Expanding $\sum_i \epsilon_i (a + x_i^T u)^3$ removes all terms involving a by (1), leaving (2).

For the distance, no weight-one Z error lies in $S^\perp$. Since the points are distinct, $e_i - e_j \in S^\perp \setminus L^\perp$ for every $i \neq j$, so $d_Z = 2$. A weight-one vector cannot lie in L: its pairing with $\mathbf{1} \in L$ under D would be nonzero. Thus $d_X \geq 2$.

For the general criterion, write $T_w(s, b, c) = \sum_i w_i s_i b_i c_i$. The difference of the physical cubic at $z + s$ and z is $3T_w(s, z, z) + 3T_w(s, s, z) + T_w(s, s, s)$. Its quadratic part must vanish as a polynomial function on L; degrees are less than p. Polarization, using $2, 3 \neq 0$, gives $T_w(s, L, L) = 0$. This condition also kills the lower terms because $s \in L$, proving sufficiency. $\square$

Watson–Campbell–Anwar–Browne [1, Definition 2 and Lemmas 6–8] impose signed orthogonality conditions that also restrict logical triple products. The proof above retains precisely the condition needed for a possibly coupled logical cubic. Related divisible-code and prime-qudit constructions appear in [2, 3, 4]. The Schur-square conductor also governs the MDS–CSS transversal-group classification [22]; here the full product $S \circ L \circ L$ matters, and the X-space is only $\langle \mathbf{1} \rangle$.

1.1 An elementary family at every prime

The following calibration example requires no conic. It supplies both the code parameters and uniqueness of the preserving cubic weight, leaving the logical phase and its resource properties as the distinctions to investigate. Let $V = \mathbb{F}_p^p / \langle \mathbf{1} \rangle$, write a_i for the image of the ith standard basis vector, and set $t = a_0$. The functional $\sigma([z]) = \sum_i z_i$ is well-defined because $p = 0$ in $\mathbb{F}_p$. Take the positive sheet to be $\{a_i\}$ and the negative sheet to be $\{a_i + t\}$.

Proposition 1.2 (Translation trade)**.** *These sheets define a strength-two trade with affine span V and $\dim L = p$. Its code is $[[2p, p-1, 2]]_p$, and its logical cubic is*

$$F(u) = -3u(t)q(u), \qquad q(u) = \sum_i u(a_i)^2 \quad (u \in V^*). \tag{3}$$

The quadratic form q has rank $p-2$. Moreover $\dim L^{\circ 2} = 2p-1$; every preserving physical cubic weight is a scalar multiple of the sheet-sign vector.

Proof. The sheets lie in the disjoint affine hyperplanes $\sigma = 1, 2$. Within a sheet the points are distinct, and their differences span $\ker \sigma$; t supplies the remaining direction. Since $\sum_i a_i = 0$, translation preserves the first two moments of this p-point set. The cubic expansion gives (3). Identify V^* with $K = \{u \in \mathbb{F}_p^p : \sum_i u_i = 0\}$. Then $q(u) = \sum_i u_i^2$; its radical on K is $\langle \mathbf{1} \rangle$, so its rank is $p-2$ and the displayed cubic is nonzero.

For rigidity, let weights α_i, β_i on the two sheets annihilate every affine quadratic. Put $\gamma = \alpha + \beta$ and $B = \sum_i \beta_i$. The linear moment says $\gamma + Be_0 \in \langle \mathbf{1} \rangle$. The quadratic moment on K has symmetric matrix

$$A = \operatorname{diag}(\gamma) + e_0\beta^T + \beta e_0^T + Be_0e_0^T.$$

A symmetric form vanishing on $K \times K$ is $\mathbf{1}r^T + r\mathbf{1}^T$ for some r. For distinct $i, j \neq 0$, comparison gives $r_i + r_j = 0$. There are at least four such indices, whence all $r_i = 0$ for $i \neq 0$. Diagonal comparison gives $\gamma_i = 0$ there; the linear moment then forces $\gamma_0 = -B$. Off-diagonal comparison gives $\beta_i = r_0$ for $i \neq 0$, and the $(0,0)$ entry gives $\beta_0 = r_0$. Thus $B = pr_0 = 0$, $\gamma = 0$, and $\alpha = -\beta$ are constant weights. The annihilator of $L^{\circ 2}$ is therefore exactly the sign line. □

1.2 The conic configurations

The matching construction has three stages. Products of secant equations give a finite list of evaluation points; their signed moments supply the code; a change of logical coordinates expresses the resulting phase through binary-form invariants. A reader interested in the resource calculation may retain Proposition 1.3 and defer the coefficient conventions to the appendix. We call these the Clebsch examples after their underlying matching geometry [20].

For the two symmetric examples take $Q = XZ - Y^2$ and label its points by $a \mapsto (1 : a : a^2)$ and $\infty \mapsto (0 : 0 : 1)$. For a perfect matching M of $\mathbb{P}^1(\mathbb{F}_p)$ define

$$P_M = \prod_{\{a,b\} \in M} L_{ab}, \quad L_{ab} = abX - (a+b)Y + Z, \quad L_{a\infty} = aX - Y.$$

Use base matchings

$$M_7 = \{02, 14, 3\infty, 56\}, \qquad M_{11} = \{01, 25, 37, 49, 68, 10\infty\}.$$

Their $\mathrm{PGL}_2(p)$ orbits have $2p$ members. The determinant square class of a transformation taking the base matching to M specifies its sign. The coefficient vector of $(P_M - P_{M_p})/Q$ specifies x_M. The exact division, sign consistency, and moment conditions are part of the finite input verification. This is the configuration studied in *Quadratic Trade Rigidity and Cubic Orientation in Conic Matching Quotients* [20].

Proposition 1.3 (Finite conic input)**.** *For the stated matchings the construction has distinct points, $\dim L = p$, and $\dim L^{\circ 2} = 2p-1$. It gives $[[14, 6, 2]]_7$ and $[[22, 10, 2]]_{11}$. In invertible logical* *coordinates, consisting of a scalar coordinate s and five quartic or nine octavic coordinates, the phases are*

$$F_7 = 4sI + 3J, \qquad F_{11} = 4sI_2 + 2I_3. \tag{4}$$

Here $I = ae - 4bd + 3c^2$ and $J = \det\left(\begin{smallmatrix} a & b & c \\ b & c & d \\ c & d & e \end{smallmatrix}\right)$ are binary-quartic invariants. For the binary octavic $f(S,T) = \sum_{i=0}^{8} \binom{8}{i} z_i S^{8-i} T^i$, $I_2 = (f,f)_8$ and $I_3 = ((f,f)_4, f)_8$ use normalized transvectants.

Finite verification. Appendix A specifies the matrices and substitutions. Exact finite-field expansion checks the moments, ranks and both polynomial identities. The signed-trade theorem then supplies the code and gate. These are finite polynomial and linear-algebra checks; the evidence record gives the independent reconstruction and its limits. □

The invariant here is the logical phase itself. This differs from using invariant theory to study weight enumerators [5]; the comparison concerns these particular binary-form phases.

2 The Hessian spectrum and product exclusion

Clifford operations permute Pauli operators up to phase. The moduli of a state's Pauli expectations therefore survive every Clifford change of frame and can obstruct conversion to independent resources. Their fourth moment will give a product-state obstruction; their rare largest nontrivial values will later obstruct a short sum-of-cubes synthesis.

For a homogeneous cubic F on $\mathbb{F}_p^k$, put $|F\rangle = p^{-k/2} \sum_u \omega^{F(u)} |u\rangle$ and $H_F(v) = (\partial_i \partial_j F)(v)$. We use natural logarithms and define

$$M_\alpha(|\psi\rangle) = \frac{1}{1-\alpha} \log \left(p^{-k} \sum_{v,w} |\langle \psi | X(v) Z(w) | \psi \rangle|^{2\alpha} \right) \quad (\alpha > 0,\ \alpha \neq 1).$$

These stabilizer Rényi entropies quantify nonstabilizerness, usually called magic, with the normalization in [6, 7]; they vanish on pure stabilizer states and are additive under tensor products. The following quadratic-sum calculation makes them accessible without enumerating all p^{2k} Paulis.

Theorem 2.1 (Rank formula)**.** *Let $p \geq 5$, and set $r(v) = \operatorname{rank} H_F(v)$ and $N_r = \#\{v : r(v) = r\}$. Then*

$$|\langle F | X(v) Z(w) | F \rangle| = \begin{cases} p^{-r(v)/2}, & w \in \operatorname{im} H_F(v), \\ 0, & \textit{otherwise}. \end{cases} \tag{5}$$

Consequently

$$M_\alpha(|F\rangle) = \frac{1}{1-\alpha} \log \left(p^{-k} \sum_{r=0}^{k} N_r p^{(1-\alpha)r} \right). \tag{6}$$

Proof. The expectation is $p^{-k} \sum_u \omega^{F(u) - F(u+v) + w^T u}$. Writing $c_v = \nabla F(v)$, the exponent, up to a constant, is $-\frac{1}{2} u^T H_F(v) u + (w - c_v)^T u$. The sum over the radical vanishes unless $w - c_v \in \operatorname{im} H_F(v)$. Euler's identity $H_F(v) v = 2c_v$ makes this equivalent to the condition in (5). On a complementary subspace diagonalize the quadratic form. Each nonzero quadratic coefficient contributes a Gauss sum of modulus $\sqrt{p}$: its squared modulus follows by replacing a pair (x,y) with $(x-y, x+y)$ in the double sum, an invertible substitution. The radical contributes $p^{k-r(v)}$, proving the modulus formula. There are $p^{r(v)}$ admissible w for each v; summing their powers proves (6). □

Kagamihara–Tsuchiya [8, Theorem 1] give the binary rank-reduction formula, with $C(x) + C(x)^T$ replacing the odd-characteristic Hessian. Chen–Yan–Zhou [9] supply related hypergraph-state moment methods. We use the established rank-reduction method in odd characteristic; the contribution below is the geometric evaluation and its resource consequences. For example, a single-qudit cubic $F(u) = u^3$ has rank zero at $u = 0$ and rank one at all other points. Thus $N_0 = 1$, $N_1 = p - 1$, and $M_2 = \log(p^2/(2p-1))$. Tensor products add these entropies. The same formula now turns the multivariate rank counts into an exact resource comparison.

Proposition 2.2 (Exact conic spectra)**.** *The nonzero entries of the vector rank distributions for the two conic phases are*

$p=7:\ r$	0	3	4	5	6
N_r	1	48	2940	26502	88158

$p=11:\ r$	0	5	6	7	8	9	10
N_r	1	120	14520	80520	21442960	2387485210	23528401270.

In each case the minimum positive rank is $(p-1)/2$, attained on exactly $p+1$ projective points. In the binary-form coordinates these are $s = 0$ and the perfect $(p-3)$rd powers. In particular,

$$M_2(F_7) = \log \frac{1977326743}{78835}, \qquad M_2(F_{11}) = \log \frac{11^{19}}{7151076991}.$$

Finite verification and deduction. The census visits vectors with first nonzero coordinate one. Since $H_F(av) = aH_F(v)$ for $a \neq 0$, every projective representative contributes $p-1$ vectors; the zero vector is added separately. The domains have 19608 and 2593742460 representatives. The $p = 7$ census has an independent implementation; the $p = 11$ exhaustive execution is checked against it in the smaller case, not independently duplicated at its full size. Substitution verifies the $p+1$ perfect-power points. The census has exactly that many minimum-rank lines, so this containment is an equality over $\mathbb{F}_p$. Equation (6) gives the entropy values. □

Lemma 2.3 (Product ceiling)**.** *For a pure state in dimension D with an orthogonal unitary Pauli basis, $M_2 \leq \log((D+1)/2)$. Thus a pure product on blocks of a and $k-a$ prime-dimensional qudits satisfies*

$$M_2 \leq \log \frac{(p^a+1)(p^{k-a}+1)}{4}. \tag{7}$$

Proof. Put $x_P = |\langle P\rangle|^2$. Purity gives $\sum_P x_P = D$ and $x_I = 1$. Cauchy–Schwarz on the $D^2 - 1$ other entries yields $\sum_P x_P^2 \geq 1 + (D-1)^2/(D^2-1) = 2D/(D+1)$. Insert this into the definition of M_2. For a product the Pauli sum factorizes, so the entropies add. □

The product-ceiling method has predecessors in the characteristic-function and qudit entropy literature [10, 7]. The bound need not be attained.

Corollary 2.4 (Clifford-product exclusion)**.** *The ten-qudit state $|F_{11}\rangle$ is not Clifford-equivalent to a product across any nontrivial bipartition. The same conclusion holds for the dihedral and Borel six-qudit conic trades over $\mathbb{F}_7$ in Appendix A. This entropy test alone does not establish the conclusion for the six-qudit Clebsch state $|F_7\rangle$.*

Proof. For fixed k, the largest right side of (7) occurs at $a = 1$ or $k-1$: the variable part $p^a + p^{k-a}$ increases toward the endpoints. At $p = 11, k = 10$ the bound is $22.6797\ldots$, below $22.8695\ldots = M_2(F_{11})$. At $p = 7, k = 6$ it is $10.4228\ldots$; the other two trades give $10.4246\ldots$ and $10.4840\ldots$, whereas $M_2(F_7) = 10.1299\ldots$. These comparisons are checked as exact rational inequalities before taking logarithms. Clifford conjugation permutes the Pauli moduli, so it preserves M_2. □

Two disjoint xyz phases at $p = 7$ have $M_2 = 8.8047\ldots$, also below F_7. The trace cubic $F(u) = \mathrm{Tr}_{\mathbb{F}_{p^k}/\mathbb{F}_p}(u^3)$ has full Hessian rank at every nonzero u, by nondegeneracy of the trace pairing, and hence $M_2 = \log(D^2/(2D-1))$, $D = p^k$. These finite-field cubic fiducials belong to the Alltop–MUB construction [11, 12, 13]. Feng–Luo [14, Proposition 2 and Appendix C] prove single-prime L1 optimality among diagonal gates. Neither that result nor the L1 ceiling of [13] is an unrestricted M_2 maximality theorem. The conjecture in [15] concerns two prime-dimensional qudits; we do not extend its attribution to arbitrary k.

3 A bounded factory comparison

The resource-count question is different from the entropy question. We compare preparing and purifying the whole six-qudit phase with purifying independent single-qudit primitives and then compiling it. This is the synthillation strategy of Campbell–Howard [16]; Li–Yeh [17, Section 4.4 and Appendix C.1] already connects signed physical qutrit gates to a coupled logical resource and its distillation. Our claim is the following finite, same-target comparison.

3.1 Noise and allowed operations

Every raw input is

$$\rho_c = (1-\delta)|M_c\rangle\langle M_c| + \frac{\delta}{6}\sum_{e\neq 0} Z(e)|M_c\rangle\langle M_c|Z(-e), \qquad |M_c\rangle = 7^{-1/2}\sum_x \omega^{cx^3}|x\rangle. \tag{8}$$

Inputs are independent; stabilizer preparation, Clifford operations, measurements, feed-forward and storage are ideal. Uniformity of the six nonzero labels is an assumption, not a consequence of diagonal twirling.

The native factory prepares $|+_L\rangle = 7^{-7/2}\sum_{z\in L}|z\rangle$, injects the fourteen signed cubic gates, measures the code stabilizers, and decodes on acceptance. Injection is deterministic: controlled subtraction from data x to a magic ancilla, followed by ancilla outcome m, supplies phase $c(x+m)^3$. A quadratic Clifford correction removes the m-dependent terms. An ancilla error $Z(e)$ becomes $Z(e)$ on the data. No injection outcome is discarded. For the resulting physical error vector e, acceptance is $\mathbf{1}^T e = 0$ and the logical error label is $E^T e$; the error is harmless exactly when $e \in L^\perp$.

The separate architecture distills each primitive independently with any finite concatenation of the three modules in Table 1, then uses deterministic linear-form cubic synthesis without final postselection. Outputs may be stored separately and successful attempts retained. Module choices and depths may differ between terms. A final Clifford is allowed if its input is a homogeneous cubic-phase state on the same six qudits. Ancilla-assisted general synthesis, pooled multi-output distillers and joint synthillation are outside this menu.

Module	points	X space	logical row	cubic weights
Weighted four-site	$0,1,2,3$	$\langle 1\rangle$	x	$(-1,3,-3,1)$
QRM six-site	$\mathbb{F}_7^*$	$\langle x\rangle$	1	all 1
RS seven-site	$\mathbb{F}_7$	$\langle 1,x\rangle$	x^2	all 1

Table 1: The three allowed one-output distillers. In each row $L = S + \langle \ell\rangle$ and the logical coefficient is -1; coordinate negation supplies $+1$. Parameters are respectively $[[4,1,2]]_7$, $[[6,1,2]]_7$, and $[[7,1,3]]_7$.

Here QRM means quantum Reed–Muller and RS means Reed–Solomon. The six-site row is the prime-qudit QRM construction [3, 19]; the length-seven RS row is checked separately. We distinguish raw supplies. In the *sign-only menu*, only $c = \pm 1$ has unit cost. The four-site module costs eight raw injections because each ± 3 gate uses three signs. Its two middle error rates are $\delta_3 = (6/7)[1 - (1 - 7\delta/6)^3]$. In the *weighted relaxation*, every nonzero cubic coefficient has unit raw cost and error δ. This relaxation favors the separate architecture.

Define the weighted cubic rank $r(F)$ as the least number of nonzero terms in $F = \sum_i c_i \ell_i^3$, with $c_i \in \mathbb{F}_p^*$ and nonzero linear forms ℓ_i over the same field. In linear-form synthesis each term consumes a primitive. The spectral calculation therefore has a second use: it limits how few independently prepared primitives can realize the target.

Lemma 3.1 (Cubic synthesis lower bound)**.** *The weighted linear-form cubic ranks obey* $9 \le r(F_7) \le 13$ *and* $15 \le r(F_{11}) \le 21$*. The rank-nine lower bound also holds for any homogeneous cubic phase on six qudits Clifford-equivalent to* $|F_7\rangle$*.*

Proof. In a shortest decomposition $F = \sum_{i=1}^r c_i \ell_i^3$, the coefficient vectors span the logical space and are projectively distinct. The Hessian at v has rank at most the number of $\ell_i(v) \neq 0$. For F_7, killing five independent forms gives $r \ge 8$. If $r = 8$, every six coefficient vectors must be independent, or a common annihilator would have Hessian rank at most two. The $\binom{8}{5} = 56$ five-subsets then span distinct hyperplanes, producing 56 minimum-rank lines. The census has only eight. Thus $r \ge 9$. For F_{11} the same argument gives $r \ge 14$ and excludes equality: $\binom{14}{9} = 2002 > 12$ minimum-rank lines would be needed. The raw signed decompositions have one zero row, giving the upper bounds. For a cubic phase the Pauli-modulus multiplicity at $p^{-r/2}$ is $N_r p^r$, so the histogram recovers the rank census. Clifford invariance therefore preserves every input to the lower-bound argument. $\square$

A weighted cube is implemented by Clifford computation of ℓ_i, one weighted cubic gate, and uncomputation. This signature-tensor synthesis framework is explicit in Heyfron–Campbell [18, Section IV]; the rank bound is for that model, not arbitrary circuits.

The comparison uses three steps. Native error detection brings the block error below the raw input error. Independence then prevents any unpurified synthesis term from meeting that target, even if errors from different terms can cancel. Finally, the rank bound forces at least nine independently purified terms. The proof retains the exact interval bounds and both raw-resource supplies.

Theorem 3.2 (Same-target resource comparison)**.** *Under* (8)*, for every* $0 < \delta \le 0.01$*, let* q *be the native factory's output block infidelity. Its expected raw consumption is less than* 16.116*. Every separate construction in the sign-only menu with block infidelity at most* q *costs at least* 54 *raw inputs per output; in the weighted relaxation it costs at least* 36*. Feasible separate constructions meeting this target exist throughout the interval.*

Proof. Write $a = 1 - \delta$ and P for native acceptance. Single errors are rejected. Every accepted weight-two error is harmful, since the evaluation points are distinct. Their probability is $(91/6)\delta^2 a^{12}$. For a fixed error support of size at least two, fixing all but its last nonzero label leaves at most one of six choices that accepts. A union bound on pairs therefore gives

$$a^{14} \le P \le 1 - 14\delta a^{13}, \qquad \frac{91\delta^2 a^{12}}{6P} \le q \le \frac{91\delta^2}{6a^{14}}. \tag{9}$$

Bernoulli's inequality implies

$$a^{12}(1 + 14\delta a) \ge (1 - 12\delta)(1 + 14\delta - 14\delta^2) = 1 + \delta(2 - 182\delta + 168\delta^2) \ge 1.$$

Hence $a^{12} \geq 1 - 14\delta a^{13} \geq P$, so $q \geq (91/6)\delta^2$, while $q/\delta \leq (91/600)/(0.99)^{14} < 1$. Also $14/P \leq 14/(0.99)^{14} < 16.116$.

A noisy primitive on a nonzero form with coefficient vector b contributes an independent error label eb. For probability distributions on the additive logical-label group, $\|\mu * \nu\|_\infty \leq \|\mu\|_\infty$, since each convolution value is an average. An unpurified raw term therefore forces final error at least δ, even with cancellations or a fixed Pauli correction. The states $Z(\eta)|F\rangle$ are orthogonal, so this probability is exactly a block-infidelity statement. An allowed final Clifford maps the ideal and errored outputs together and preserves their overlaps. Since $q < \delta$, all synthesis terms must be purified. Each independent tree has at least six raw leaves in the sign-only menu and four in the weighted menu. Retrying cannot lower those counts. Lemma 3.1 requires at least nine terms, giving 54 and 36.

For feasibility use the thirteen-term signed decomposition. A weighted four-site primitive has error at most δ^2/a^4, so thirteen give block error at most $13\delta^2/(0.99)^4 < (91/6)\delta^2$. For the seven-site module harmful accepted errors have weight at least three, giving error at most $35\delta^3/a^7$; thirteen give $455\delta^3/a^7 < (91/6)\delta^2$ throughout the interval. A union bound suffices for these feasible upper estimates; the lower bound already allowed cancellations. $\square$

At $\delta = 0.01$ the exact enumerator calculation gives the comparison below. The separate rows are feasible constructions, not claims to attain their menu's optimum.

Construction	block infidelity	expected raw inputs
Native fourteen-input	0.00159853	16.0894
Weighted four-site, thirteen terms	0.00133141	54.1275
Sign-only seven-site, thirteen terms	0.0000130920	97.6325

For an ordinary enumerator A_w of $L^\perp$, native good-output probability before conditioning is $\sum_w A_w a^{14-w}(\delta/6)^w$. Acceptance also has the closed form $P = [1 + 6(1 - 7\delta/6)^{14}]/7$. Thus the calculation includes harmless accepted errors, rather than treating every nonzero physical error as failure.

4 A chordal restriction of the logical cubic

The logical cubic also provides an interface with the invariant pencil in *Chordal and Conference Cubics: Reconstruction and a Residual C_2-Torsor* [21]. A restriction of a logical polynomial is a resource on fewer variables; it does not by itself construct a subcode. The useful question is whether this smaller resource retains a distinction that the Clifford-invariant spectrum can detect.

Write $A = \{y \in \mathbb{F}_{11}^6 : \sum_i y_i = 0\}$ for the five-dimensional augmentation space of the six matching pairs, ordered as

$$(\{0,1\}, \{10,\infty\}, \{2,5\}, \{3,7\}, \{4,9\}, \{6,8\}).$$

The geometric construction supplies an embedding $\psi : A \longrightarrow V_5$ and two A_5-invariant cubics: the restriction $h = F \circ \psi$ and a conference cubic c. Here a conference cubic is obtained from the signed triangle sum $\sum_{i<j<k} B_{ij}B_{jk}B_{ki}y_iy_jy_k$, restricted to A, where B is the chosen symmetric conference matrix: its diagonal is zero, its off-diagonal entries are ± 1, and $B^2 = 5I$. The chosen signing has A_5-invariant triangle products. The coefficient certificate uses the basis $e_i - e_5$ $(0 \leq i < 5)$ of A. A chordal cubic is the secant hypersurface of a rational normal quartic; the Hankel determinant below is its standard equation. The span of h and c is the invariant pencil, a two-dimensional space of cubics considered up to nonzero scalar. The supplied coefficient certificate fixes actual representatives, since scalar multiples can define different phase functions in fixed coordinates.

Proposition 4.1 (Chordal shadow and sign choices)**.** *Over $\mathbb{F}_{11}$ the stabilizer A_5 of the base matching acts on the logical space as $\mathbf{1}\oplus V_4\oplus V_5$. The restriction to its unique five-dimensional constituent is, in the recorded coordinates,*

$$8\det\begin{pmatrix} v_0 & v_1 & v_2 \\ v_1 & v_2 & v_3 \\ v_2 & v_3 & v_4 \end{pmatrix}.$$

Here $v_0,\dots,v_4$ are new linear coordinates on A. The restriction is a chordal member of the invariant pencil. At ranks $0,3,4,5$, the chordal and conference censuses are respectively

$$(1, 120, 27720, 133210), \qquad (1, 300, 22260, 138490).$$

Thus their five-qudit phase states are not Clifford-equivalent. At $p = 7$ the restriction $s = 0$ in (4) *is $3J$. Reversing the sheets sends U_F to U_F^{-1}, a Clifford-conjugate gate. Let q be the involution on A induced by the permutation* $(01)(24)(35)$ *of these ordered pairs. With actual representatives h and $h_2 = h \circ q$, q exchanges the two chordal members and gives a linear Clifford equivalence of their five-qudit states.*

Finite identification and operational deduction. Character idempotents in the reconstructed A_5 action have ranks $1, 4, 5$. The recorded intertwiner identifies V_5 with the augmentation module on the six matching pairs; polynomial substitution gives the Hankel determinant. The two exact censuses then separate the resources by Theorem 2.1. The $p = 7$ restriction follows directly from (4). Sheet reversal changes every sign in (2), hence negates F. For an invertible linear map q, the basis permutation $|z\rangle \mapsto |qz\rangle$ is Clifford: it sends $X(v)$ to $X(qv)$ and $Z(w)$ to $Z(q^{-T}w)$. In the coefficient certificate, $h \circ q = 2\widehat{h}_2$, where $\widehat{h}_2$ is the projectively normalized second chordal cubic. Taking $h_2 = h \circ q$ gives $h_2 \circ q = h$ because $q^2 = I$; the basis permutation for q therefore exchanges $|h\rangle$ and $|h_2\rangle$. If normalized representatives are preferred, the basis permutation for $7q$ sends $|h\rangle$ to $|\widehat{h}_2\rangle$: its inverse is $8q$, and $2 \cdot 8^3 = 1$ in $\mathbb{F}_{11}$. For sheet reversal, homogeneity gives $F(-u) = -F(u)$, so coordinate negation conjugates U_F to U_F^{-1} and sends $|F\rangle$ to $|-F\rangle$. The q-exchange identity is on the shadow, not a claim about all extensions to the full ten-qudit gate. □

The singular scheme of the Hankel cubic is a rational normal quartic; its twelve $\mathbb{F}_{11}$-points are not the entire scheme. The existing lift test rules out the identity-on-complement extension. For geometric lifts, checking all 1320 elements of $\mathrm{PGL}_2(11)$ gives exactly 60 that preserve V_5, namely the reconstructed A_5. Their restrictions have determinant 1, whereas $\det q = -1$ on A. Thus none restricts to q. This excludes the specified geometric lifts; it does not classify arbitrary full-gate lifts.

Questions suggested by the construction

The translation family shows that the parameters and cubic-weight rigidity alone do not select the symmetric examples. Their distinction lies in the phase geometry and its spectrum. The conic translation-class search finds no examples at $p = 17$ or 19, after finding examples at 7, 11 and 13. Does this finite pattern reflect an obstruction at larger primes, or do examples recur? A second question is whether a balanced $2p$-point trade can realize a trace-cubic resource on $p-1$ logical qudits. Finally, the distance-three search inside the fixed $p = 7$ evaluation space fails, while the elementary distance-three RS examples do not establish the same product exclusion. Finding a distance-at-least-three code with several logical qudits and such an invariant phase is a separate construction problem. The present factory comparison leaves pooled distillation and unrestricted adaptive preparation open.

A Finite data and verification boundary

The arguments proving the signed-moment dictionary, translation family, Hessian formula, product ceiling and interval comparison are given in the text. The conic matrices, polynomial identities, exact spectra and shadow identification use finite computations. No Lean coverage is claimed for this companion. The source annotations identify each dependence; the claim map pins every theorem statement and records formal coverage as absent.

A.1 Coordinates and exact inputs

The evidence index supplies the full matrices, signs and raw cubics from the committed reconstruction bundle. For $p = 7$ the coefficient order is $X^2, XY, XZ, Y^2, YZ, Z^2$ and the logical substitution is $u = (a, b, s + c, 3s + c, d, e)$. For $p = 11$ the selected coefficient order is

$$X^4, X^3Y, X^3Z, X^2YZ, X^2Z^2, XY^2Z, XYZ^2, XZ^3, YZ^3, Z^4;$$

the substitution is $u = (z_0, z_1, 7z_2, 6z_3, 3s + 6z_4, s + 3z_4, 6z_5, 7z_6, z_7, z_8)$. The transvectant normalization is

$$(f, g)_r = \frac{(m-r)!(n-r)!}{m!n!} \sum_{j=0}^{r} (-1)^j \binom{r}{j} \frac{\partial^r f}{\partial S^{r-j} \partial T^j} \frac{\partial^r g}{\partial S^j \partial T^{r-j}},$$

for binary forms of degrees m, n; all denominators used here are invertible.

For these matching configurations, the signed cubic line is also the Macaulay inverse system of the Artinian reduction with Hilbert function $(1, p-1, p-1, 1)$ [20, Corollary on self-association and cubic duality]. Thus the same cubic enters the geometric duality and the linear-form compilation problem.

For the other $p = 7$ trades, one may reconstruct the positive and negative sheets as translation classes of the following matchings. Entries are partner lists on $0, \ldots, 6, \infty$, with index 7 representing infinity:

	$+$	$-$
dihedral	$(1, 0, 5, 4, 3, 2, 7, 6)$	$(1, 0, 6, 4, 3, 7, 2, 5)$
Borel	$(1, 0, 4, 6, 2, 7, 3, 5)$	$(1, 0, 6, 5, 7, 3, 2, 4)$.

Their vector censuses at ranks $0, 3, 4, 5, 6$ are respectively

$$(1, 6, 1092, 20202, 96348), \qquad (1, 6, 588, 20286, 96768).$$

Their fourth-moment sums are $58711/16807$ and $55327/16807$.

A.2 Classification and distance computations

The conic search enumerates perfect matchings, identifies their translation classes, and buckets classes by equal first and second moment tensors. Pairs in a bucket are tested for a nonzero third-moment difference, then quotiented by the stated affine action. The respective numbers of translation classes for $p = 5, 7, 11, 13, 17, 19$ are $3, 15, 945, 10395, 2027025, 34459425$; the numbers of inequivalent hits under $\mathrm{AGL}(1, p)$ are $0, 3, 2, 1, 0, 0$. This is a finite source classification, not a classification of all trades or all Clifford-equivalent codes. Every hit has the signed moment condition, $\dim L = p$ and Schur-square dimension $2p - 1$. The $p = 13$ hit gives $[[26, 12, 2]]_{13}$; its stored 20000-vector Hessian sample is not an exhaustive spectrum or a proved minimum-rank bound. The new $p = 11$ and $p = 13$ X-distance estimates are only upper bounds.

The fixed-space distance search uses all $L' \subseteq L$ containing $\mathbf{1}$ with $\dim L' \geq 3$. Put $R(L') = \{s \in L' : T(s, L', L') = 0\}$. It suffices to test the maximal admissible X-space $R(L')$. A weight-two $e_i - e_j$ is undetected exactly when columns i, j of this space agree; it is harmless exactly when those columns of L' agree. Thus distance at least three requires equality of the two column partitions. Reduced row-echelon enumeration of subspaces of $L/\langle \mathbf{1} \rangle$ visits 61927311 candidates of dimensions two through six. None with a nonzero restricted cubic passes the partition test. The one-dimensional case also fails: a nonzero one-variable cubic has zero trilinear radical on that quotient, so its only available X-space is $\langle \mathbf{1} \rangle$, which does not separate its distinct evaluation columns. This excludes the specified nonzero-cubic subcodes inside the fixed L; it does not exclude other evaluation spaces, concatenation or code switching. For comparison, evaluation on all of $\mathbb{F}_p$ with X-space RS_2 gives the length-p distance-three rows $[[7,1,3]]_7$, $[[11,3,3]]_{11}$ and $[[13,4,3]]_{13}$ in the recorded polynomial-code search. They are distinct from Campbell's length-$(p-1)$ construction [19]; their checks belong to the strengthening bundle. The exhaustive negative has an internal count check and recorded tally, but no independent full second implementation.

A.3 Artifacts and reproducibility

The machine-readable evidence registry in `verification/evidence.json` records the exact repository paths, checksums, byte counts, replay commands, search domains and limits. The inputs are the committed reconstruction, spectrum, strengthening and shadow bundles, together with the factory benchmark. The factory certificate contains all native and primitive enumerators, exact rational sample values and interval constants. Its checks compare primal enumeration with MacWilliams transforms and synthesis-kernel enumeration with the Fourier expression; no random seed or floating-point decision enters that benchmark.

The lightweight draft check verifies statement identities, annotation links, input hashes, census arithmetic and the exact product and interval inequalities. A fresh standard-library check also re-expands both coordinate normal forms, checks both conic moment and Schur-square identities, reproduces the complete $p = 7$ census, and checks seven primes in the translation family. A separate check reproduces the actual shadow normalization and compares all 1320 geometric actions with the residual involution. It does not rerun the multi-billion-point $p = 11$ census or the matching exhaustions. Those remain explicitly attributed to the recorded executions; full commands and their runtime estimates accompany the evidence registry. The accompanying supplement contains the inputs and programs needed for these checks. The lightweight check does not independently reproduce the large searches.

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